

If λ doubles, ΔE decrease by 2 times.

29 fastest way to solve is via trial and error of each option:

${}_{90}^{232}\text{Th}$ (6 alpha decays)

$\rightarrow {}_{78}^{208}\text{Pt}$ (4 beta decays)

$\rightarrow {}_{82}^{208}\text{Pb}$

Cannot tell from mass number and proton number whether gamma decay takes place or not because gamma decay do not change any of the above numbers.

20 Let n be the number of half lives that would have passed when the 500 stockpile of bomb is